

KALEEM AHMED

+91 6306026859 ◇ Noida, India

Portfolio ◇ shaamidreez@gmail.com ◇ [LinkedIn](#) ◇ [GitHub](#) ◇ [LeetCode](#)

SUMMARY

Backend-focused Software Engineer building distributed systems in Node.js, TypeScript and Python/FastAPI: microservices, event-driven pipelines, idempotent payment flows and production LLM/RAG features, owned end to end from service boundaries to deployment and observability.

SKILLS

Languages & Backend: Node.js, Express.js, TypeScript, JavaScript, React/Next, Python, REST API Design

Data & Messaging: MongoDB, PostgreSQL, Prisma, SQL, Redis, RabbitMQ, Message Queues

AI/LLM Engineering: LLM-Powered Applications, RAG, Vector Search, Prompt & Context Engineering, Agentic Workflows, OCR, Spec-Driven Development, Human-in-the-Loop Guardrails

Real-Time & Frontend: Socket.IO, WebRTC, WebSockets, Next.js, React.js, Tailwind CSS

DevOps & Observability: Docker, AWS (IAM, EC2, Lambda, S3, VPC), Nginx, Prometheus, Grafana, Winston

Integrations & Payments: Razorpay, Shiprocket, Stripe, Salesforce REST & Metadata APIs, WhatsApp API

EXPERIENCE

Software Developer — Astrea IT Services

Apr 2024 – Present

Backend, AI and full-stack delivery on client platforms using Node.js, Python, Docker, AWS, Next.js and Salesforce

- **AI Document Automation (Primble):** Contributed to Primble, platform that fills **Insurance forms** (up to 1,135 fields each) from an uploaded policy PDF — Google Vision OCR, LLM fact extraction, then a **deterministic-first hybrid fill** where rule and alias passes cover most fields at zero token cost and a single combined LLM pass handles only the true gaps across every selected form at once, Quality Scoring, AI suggestions, form generation for missing fields. **Evidence-gated answers** and type-aware rejection enforce a strict *blank-over-wrong* rule on a legal document. Cut manual data entry **80%+ per submission**.
- **Data Synchronization ETL Hub:** Built a Node.js middleware server keeping a file-based **SAP** system, **Salesforce** and an ecommerce platform consistent in both directions across **14 business object types**, cutting manual reconciliation an estimated **70–75%**. An embedded FTP server and file watcher ingest SAP JSON into async pipelines with **auto-discovered routing**, so a new integration is a single file. Implemented source-aware **idempotent UPSERTs on composite keys** with transaction boundaries, OAuth2 token caching, retry with backoff, and failures written back to Salesforce as records.
- **ECOU — Voice-First B2B Ordering Platform (client):** Built a voice-driven ordering and dispatch platform for a Romanian agriculture client. **Gemini native-audio** transcription detects the spoken language and a **grounded extraction** step resolves product, quantity, unit and location against catalogue IDs, so the model cannot invent entities. Added deterministic unit conversion, bounded spoken clarifications, and a **human confirmation gate** before dispatch; matched orders fan out over **WhatsApp Cloud API and Twilio** behind one provider interface. Next.js 16, PostgreSQL, Prisma, bilingual EN/RO, plus an admin back office for catalogue, pricing and stock.
- **Delivery and ownership:** Took features from requirement to production across three client platforms — writing the spec, designing the data model and service boundaries, building both halves and deploying them. Dockerised services behind Nginx on AWS, CI/CD pipelines, structured logging and Sentry error tracking, and handover documentation on each system, since none of them stays with its author.

AI ENGINEERING & ADAPTABILITY

Actively practicing AI-assisted development — AI as a force multiplier, with full engineering ownership and human-in-the-loop oversight, and judgment applied to every output by reviewing, challenging and overriding it. The durable advantages are still human: focus, discipline, adaptive learning and product taste.

Tools & Focus: OpenAI Codex, Claude, Cursor, context management, N8N, AI-assisted debugging, engineering automation.

PROJECTS

[User](#) | [Seller](#) | [Admin](#) | [Test Login](#)*Next.js, TypeScript, Express.js, MongoDB, Prisma, Redis, RabbitMQ, Docker, AWS EC2, Nginx, Razorpay, Shiprocket, Socket.IO*

- **Built as the sole engineer** — a multi-vendor marketplace driven by a **seller-authored customization schema** (typed fields, validation rules, reusable templates); a purchase is invalid until the buyer satisfies it, re-validated at checkout so a stale tab cannot smuggle an edited payload past payment.
- **12 backend microservices and 3 independently deployed Next.js apps.** Orders are never created in the checkout request: a signed **Razorpay webhook** publishes an event and **idempotent RabbitMQ consumers** drive order lifecycle, wallet accounting, inventory sync, refunds and payouts, with retries and **dead-letter queues**. Immutable wallet ledgers with atomic conditional updates closing a TOCTOU race found in a self-audit, delivery-gated payouts, JWT-based RBAC, Redis caching, rate limiting and **Prometheus/Grafana/Loki** observability.
- **Full commerce surface:** multi-vendor storefronts, search and shop reviews; a cart treating two differently-customized units of the same product as separate lines; server-recomputed pricing, stock and coupons at checkout; **Shiprocket** shipping behind a circuit breaker with AWB generation and webhook-driven tracking; seller verification, moderation and Socket.IO chat.

Forever Yours (Shaza) — Real-Time Couples Platform with AI Companion[Live App](#)*Node.js, Express.js, Socket.IO, WebRTC, Next.js, MongoDB Atlas Vector Search, Redis, Jina Embeddings, Groq/Gemini*

- **WebRTC** voice and video calling with a host-authoritative synced watch-together room over Socket.IO signalling, handling ICE-candidate buffering, call glare, presence and reconnect recovery.
- **Production RAG companion** learning a partner's texting style from imported chat exports over **MongoDB Atlas Vector Search**, using a composite retrieval score (relevance dominant, recency and importance bounded), **reliability-tiered retrieval** and a fact-check protocol that refuses to fabricate memories; a paired AI mediator's **map-reduce design structurally isolates** each partner's private input.
- **Product surface: Watch Together** synced playback, a collaborative canvas on Liveblocks/Yjs, memories with map and graph views, scheduled letters and cron-unlocked gifts. A **mood tracker** of 8 moods, 32 sub-moods and 12 needs tags, with per-field privacy enforced at a single projection choke point, streaks, cycle tracking and async AI summaries off the save path. An **AI Clash Resolver** running a five-state machine with two-party consent on every transition that matters, plus a private coach, a breakup-discernment path, and HMAC Razorpay subscriptions.

Gaza40+ Student Portal — Humanitarian Operations Platform[Live App](#)*Next.js, Express.js, TypeScript, MongoDB, Prisma, Socket.IO, Cloudflare R2, AWS SDK, JWT, Tailwind*

- Humanitarian operations platform **serving 500+ DAUs**, replacing coordination across Notion, WhatsApp and spreadsheets with one system for onboarding, offers, documents and case lifecycle.
- **Layered authorization model** — 5-tier RBAC with a canonical **scope resolver** returning a discriminated union, case region derived from the university's country rather than the user's profile, and document-type allowlists held in code so a database write cannot grant access. Configurable visa and financial-gap engine computed on read, real-time chat, and **Cloudflare R2** documents streamed through the API.
- **Operational workflows:** country-routed query triage with routing held in configuration rather than an enum, where **escalation releases the assignment** and counts how many people it reached before declaring success; offer review with normalized before/after **revision snapshots** and a no-op return when nothing changed; and background **CSV export jobs**, capped per user, deduplicated on repeated filters within five minutes, columns allowlisted, and the authorization scope frozen at creation.

EDUCATION**B.Tech in Computer Science Engineering** — Abdul Kalam Technical University, CGPA 8.3/10 2020 – 2024

ACHIEVEMENTS

- Solved **1500+ DSA problems** across LeetCode and GeeksforGeeks; **peak LeetCode rating 1798, Global Rank 812** in Weekly Contest 359.
- **Global Rank 800** in TCS CodeVita Season 11 among **110,000+ participants** worldwide.